

TB Group Storage Inventory & Catalog

Comprehensive Asset Audit, Multi-Omics Modality Breakdown & Data Governance Assessment for /work/TB_group/

PLATFORM & WORKSPACE
SDU eScience UCloud • BINF INFIMM

WEKAFS MOUNT POINT
/work/TB_group/ (Folder #1)

SURVEYED ASSETS
29 Datasets • 957,970 Files

SURVEY DATE & SCOPE
September 2026 • 2023–2026 Cohorts

TOTAL STORAGE FOOTPRINT5.51 TB
5,643.01 GB
Active persistent working volume on WekaFS 4.7 PB parallel tier

FILE OBJECT COUNT957.9k files
957,970
High directory density: 268,358 nested subdirectories

PRIMARY RAW DATA67.8% volume
3,826.48 GB
14 datasets: Raw FASTQs (1.82 TB) & Zeiss/Leica slides (2.00 TB)

ANALYSIS & PROCESSED32.2% volume
1,816.53 GB
15 datasets: Single-cell H5/RDS matrices & QuPath annotations

GIT REPOSITORIES11 tracked
11 directories
Version-controlled pipeline code, analysis repos & notebooks

SNAPSHOT BACKUP STATUS15 Backed Up
15 / 29
15 datasets with backup copies; 14 require Restic snapshot registration

Storage Allocation by Research Modality

Microscopy (Raw)

Histology (QuPath)

Single-Cell Objects

35.5% (2002.7 GB • 11 datasets)

20.0% (1127.2 GB • 5 datasets)

1.9% (109.6 GB • 2 datasets)

Raw Sequencing (FASTQ)

Single-Cell Project

Code / Pipeline

32.3% (1823.8 GB • 3 datasets)

10.3% (579.8 GB • 7 datasets)

<0.1% (0.0 GB • 1 datasets)

Modality Classification	Datasets	Volume (GB)	% Share	Total Files
Microscopy & Raw Imaging Data	11	2002.65 GB	35.5%	678,848
Raw Sequencing (FASTQ)	3	1823.83 GB	32.3%	264
Histology & Image Analysis (QuPath)	5	1127.16 GB	20.0%	12,031
Single-Cell Analysis Project	7	579.79 GB	10.3%	178,298
Single-Cell Processed Objects	2	109.58 GB	1.9%	88,525
Code / Analysis Pipeline	1	0.00 GB	0.0%	4

- Executive Findings & Governance Priorities
- Raw Data Preservation (Tier 1 Priority):** 3,826.48 GB across 14 directories constitutes irreplaceable raw FASTQs and Zeiss CZI slides. These folders must be write-protected (read-only POSIX permissions) and registered in the automated incremental Restic backup to /work/Data_backup/repo/.
 - Git Remote Synchronization:** 11 directories contain .git repositories (e.g. TB_TL, TB-Endothelial, tb-lung-atlas). Verify that all local commits and branches are pushed to GitHub, and ensure large intermediate files (H5, RDS) are excluded via .gitignore.
 - Storage Deduplication Opportunity:** QuPath Analysis F2944(1) (0.02 GB, 707 files) is a redundant clone of QuPath Analysis F2944 (0.51 GB). Verify uniqueness and remove the clone to maintain project hygiene.
 - HPC Job Mounting Protocol:** Always mount TB_group as **Folder #1** and Data_backup as **Folder #2** on UCloud. Never leave unlinked deliverables in container root /work/.

Master Directory Catalog — Part I (Datasets 01 to 15)

Detailed breakdown of primary datasets, sample markers, technical stacks, storage footprints, and suggested governance actions.

#	DIRECTORY & METADATA	MODALITY / TYPE	SIZE & FILES	SAMPLE IDS / MARKERS	TOP FORMATS	FLAGS	GOVERNANCE ACTION
1	NAFP Image analysis 2025-01 to 2026-08 • 2,558 dirs	Histology (QuPath)	9.51 GB 9,709 files	CD31 CD312 CD313	.json (4,794) .jpg (2,409) .qpdata (2,371)	BAK	ROUTINE SNAPSHOT Maintain regular incremental snapshots in Data_backup.
2	QuPath Analysis F2944 2025-01 to 2025-02 • 188 dirs	Histology (QuPath)	0.51 GB 735 files	F2944	.json (363) .qpdata (184) .jpg (184)	BAK	ROUTINE SNAPSHOT Maintain regular incremental snapshots in Data_backup.
3	QuPath Analysis F2944(1) 2025-01 to 2025-01 • 178 dirs	Histology (QuPath)	0.02 GB 707 files	F2944	.json (353) .jpg (176) .qpdata (176)	BAK	REDUNDANCY AUDIT Compare against master project; verify uniqueness & prune.
4	TB-Endothelial 2024-04 to 2026-03 • 3,473 dirs	Single-Cell Project	0.86 GB 24,239 files	CD86 Endothelial cd3679 f0098 +4	.hpp (13,231) .rds (1,855) (no ext) (1,649)	GIT DOC VENV IPYNB BAK	GIT REMOTE SYNC Verify local commits and tags pushed to remote GitHub repo.
5	TB-VIT 2024-04 to 2026-09 • 1,835 dirs	Single-Cell Project	10.08 GB 7,700 files	CD0 CD8744 Cd27 F2662 +4	(no ext) (2,300) .r (747) .rds (742)	GIT DOC VENV IPYNB BAK	GIT REMOTE SYNC Verify local commits and tags pushed to remote GitHub repo.
6	TB-scverse 2019-10 to 2026-09 • 3,726 dirs	Microscopy (Raw)	20.69 GB 29,388 files	CD103 CD4 Cd103 cd0 +4	.py (9,443) .pyc (9,441) (no ext) (2,354)	GIT DOC VENV IPYNB BAK	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
7	TB-total-lung 2023-08 to 2026-09 • 8,546 dirs	Microscopy (Raw)	25.82 GB 54,219 files	CD077815 CD7 Cd27 F0806 +4	.hpp (25,862) (no ext) (4,454) .rds (4,394)	GIT DOC VENV IPYNB BAK	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
8	TB_TL 2025-09 to 2026-08 • 237 dirs	Single-Cell Project	173.60 GB 770 files	Cd19 Cd226 Cd3 Cd4 +4	.png (403) (no ext) (183) .csv (66)	GIT DOC IPYNB	GIT REMOTE SYNC Verify local commits and tags pushed to remote GitHub repo.
9	TB_TL_AdTx_F2998_F3035 2026-05 to 2026-07 • 82,912 dirs	Single-Cell Project	236.15 GB 104,286 files	AdTx CD4 F2998 F3035 +4	(no ext) (98,999) .csv (3,068) .h5 (650)	GIT DOC IPYNB BAK	GIT REMOTE SYNC Verify local commits and tags pushed to remote GitHub repo.
10	TB_TL_Non_Tcell 2025-02 to 2026-08 • 220 dirs	Single-Cell Project	123.54 GB 824 files	Cd28 Cd3 Cd40 cd0 +4	.png (436) (no ext) (249) .pdf (58)	GIT DOC IPYNB BAK	GIT REMOTE SYNC Verify local commits and tags pushed to remote GitHub repo.
11	TB_TL_TBD2_F2998 2026-04 to 2026-04 • 38,232 dirs	Single-Cell Objects	55.12 GB 44,192 files	CD4 F2998 F3035 TBD2	(no ext) (41,585) .csv (1,549) .h5 (326)	—	ROUTINE SNAPSHOT Maintain regular incremental snapshots in Data_backup.
12	TB_TL_TBD3_F2998_F3035 2026-05 to 2026-05 • 38,274 dirs	Single-Cell Objects	54.46 GB 44,333 files	CD4 F2998 F3035 TBD3	(no ext) (41,683) .csv (1,549) .h5 (326)	—	ROUTINE SNAPSHOT Maintain regular incremental snapshots in Data_backup.
13	TB_TL_TBD_F2998 2026-02 to 2026-04 • 20,773 dirs	Single-Cell Project	34.87 GB 24,638 files	CD4 F2998 TBD f5664	(no ext) (23,222) .csv (821) .h5 (166)	GIT DOC IPYNB	GIT REMOTE SYNC Verify local commits and tags pushed to remote GitHub repo.
14	imaging_analysis_f2805_HEV 2016-09 to 2026-06 • 15,489 dirs	Microscopy (Raw)	134.32 GB 150,254 files	F2805 HEV cd00 cd01 +4	.pyc (49,690) .py (49,689) .h (18,814)	DOC VENV IPYNB BAK	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
15	imaging_data_CD31pilot_#1 #2 2024-07 to 2024-07 • 0 dirs	Microscopy (Raw)	83.57 GB 5 files	CD31	.czi (5)	—	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.

Master Directory Catalog — Part II (Datasets 16 to 29)

High-volume imaging runs, raw sequencing archives, and single-cell atlas workflows.

#	DIRECTORY & METADATA	MODALITY / TYPE	SIZE & FILES	SAMPLE IDS / MARKERS	TOP FORMATS	FLAGS	GOVERNANCE ACTION
16	 imaging_data_f2630_HEV 2024-06 to 2024-06 • 0 dirs	Microscopy (Raw)	260.63 GB 20 files	HEV f2630	.czi (20)	—	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
17	 imaging_data_f2630_neutrophils 2024-05 to 2024-08 • 36 dirs	Histology (QuPath)	256.74 GB 158 files	f2630 neutrophil	.json (61) .jpg (33) .czi (32)	BAK	ROUTINE SNAPSHOT Maintain regular incremental snapshots in Data_backup.
18	 imaging_data_f2701_HEV 2023-07 to 2024-06 • 2 dirs	Microscopy (Raw)	228.16 GB 26 files	F2805 HEV f2701	.czi (26)	—	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
19	 imaging_data_f2805_HEV 2023-07 to 2023-07 • 0 dirs	Microscopy (Raw)	53.57 GB 19 files	F2805 HEV f2805	.czi (19)	BAK	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
20	 imaging_data_f2944_1 2024-10 to 2025-01 • 177 dirs	Histology (QuPath)	860.38 GB 722 files	f2944	.json (349) .jpg (174) .qpdata (174)	BAK	ROUTINE SNAPSHOT Maintain regular incremental snapshots in Data_backup.
21	 imaging_data_f2944_2 2024-11 to 2024-11 • 0 dirs	Microscopy (Raw)	718.16 GB 21 files	f2944	.czi (21)	—	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
22	 imaging_data_seattle 2023-11 to 2023-11 • 0 dirs	Microscopy (Raw)	22.15 GB 1 files	—	.lif (1)	—	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
23	 imaging_data_test_cellpose 2022-04 to 2026-08 • 154 dirs	Microscopy (Raw)	244.82 GB 11,301 files	CD31 cd0 cd1398 cd3 +4	.tif (7,218) .npy (3,863) (no ext) (122)	GIT DOC IPYNB	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
24	 imaging_py 2024-07 to 2024-07 • 1 dirs	Code / Pipeline	0.00 GB 4 files	—	.ipynb (4)	IPYNB	ROUTINE SNAPSHOT Maintain regular incremental snapshots in Data_backup.
25	 nagar_F2657_csv 2006-05 to 2026-08 • 49,714 dirs	Microscopy (Raw)	210.76 GB 433,594 files	CD27 CD4 F2000 F2008 +4	.py (110,083) .pyc (100,574) .h (66,203)	GIT DOC IPYNB BAK	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
26	 sequencing_data_MonkeyPAXgene_GE0241235 2023-06 to 2023-08 • 0 dirs	Raw Sequencing (FASTQ)	144.49 GB 62 files	Monkey PAXgene	.gz (60) .tsv (1) .xlsx (1)	—	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
27	 sequencing_data_endothelial 2023-12 to 2025-03 • 0 dirs	Raw Sequencing (FASTQ)	670.41 GB 138 files	endothelial	.gz (136) .md5 (1) .eve8n4 (1)	—	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
28	 sequencing_data_totallung 2024-01 to 2024-01 • 0 dirs	Raw Sequencing (FASTQ)	1008.93 GB 64 files	totallung	.gz (64)	—	PRIMARY RAW BACKUP Write-protect; register in automated Restic backup to Data_backup.
29	 tb-lung-atlas 2024-04 to 2025-03 • 1,633 dirs	Single-Cell Project	0.69 GB 15,841 files	cd09 cd27 cd5 f0409 +4	.hpp (12,916) (no ext) (555) .r (513)	GIT DOC VENV BAK	GIT REMOTE SYNC Verify local commits and tags pushed to remote GitHub repo.

TOTAL SURVEYED ITEMS

29 Directories

TOTAL STORAGE FOOTPRINT

5,643.01 GB (5.51 TB)

TOTAL FILE INVENTORY

957,970 Files

TOTAL DIRECTORY TREES

268,358 Folders

Primary Raw Backup

14 Datasets • 3,826.5 GB

Immutable raw FASTQs and microscopy scans. Priority Restic backup snapshots required.

Git Remote Sync

8 Repos • 570.7 GB

Single-cell and analysis code repos. Verify local commits and tags pushed to GitHub.

Routine Snapshot

6 Datasets • 1,245.8 GB

Processed objects and QuPath workspaces. Maintain scheduled incremental backup.

Redundancy Audit

1 Dataset • 0.02 GB

Duplicate QuPath folder candidate. Verify uniqueness against master project and prune.

Granular technical review of sequencing formats, high-resolution microscopy slides, digital pathology, and single-cell pipelines.

1,823.83 GB • 32.3%

3 cohorts 264 files .fastq.gz, .md5, .tsv

- sequencing_data_totallung: 1,008.93 GB (64 files) • Demultiplexed pooled runs (I1, I2, R1, R2).
- sequencing_data_endothelial: 670.41 GB (138 files) • Endothelial sorted paired libraries.
- sequencing_data_MonkeyPAXgene_GEO241235: 144.49 GB (62 files) • Public benchmark RNA-seq.

Governance Requirement: Raw FASTQs are strictly immutable. Write-protect POSIX directory permissions. Ensure source MD5 verification checksums are archived alongside data.

2,002.65 GB • 35.5%

11 cohorts 678,848 files .dzi, .lif, .tif, .npy

- **Zeiss Slide Scans:** f2944_2 (718.2 GB), f2630_HEV (260.6 GB), f2701_HEV (228.2 GB), f2805_HEV (53.6 GB), CD3ipilot (83.6 GB).
- **Leica Slides:** imaging_data_seattle (22.15 GB .lif).
- **Cellpose & Spatial ML:** imaging_data_test_cellpose (244.8 GB), nagar_F2657_csv (210.8 GB, 433k files - SIM testing, QUICHE).

Storage Note: Slide files (.czi/.lif) average 5–40 GB per slide. Avoid repeated unpacking or decompression on WekaFS. Register in Restic incremental snapshots.

1,127.16 GB • 20.0%

5 projects 12,031 files .qpdata, .qpproj, .json, .csv

- **imaging_data_f2944_1:** 860.38 GB (722 files) • Major QuPath project with embedded raw CZI references.
- **imaging_data_f2630_neutrophils:** 256.74 GB (158 files) • Neutrophil spatial infiltrate quantification.
- **NAFP Image analysis:** 9.51 GB (9,709 files) • Multi-cohort HEV vs iBALT annotations.
- **QuPath Analysis F2944** (510 MB) vs **QuPath Analysis F2944(1)** (20 MB).

Redundancy Alert: QuPath Analysis F2944(1) is an unlinked duplicate.
Conduct differential check against main project and safely prune to reclaim inodes.

689.37 GB • 12.2%

9 projects **266,823 files** **.rds, .h5, .h5ad, .r, .py**

- **10x Matrices:** TB_TL_AdTx_F2998_F3035 (236.2 GB, 104k files), TB_TL (173.6 GB), TB_TL_Non_Tcell (123.5 GB).
- **Processed RDS Objects:** TB_TL_TBD2_F2998 (55.1 GB) • TB_TL_TBD3_F2998_F3035 (54.5 GB) • TB_TL_TBD_F2998 (34.9 GB).
- **Reproducible Git Pipelines:** TB-Endothelial, TB-VIT, tb-lung-atlas.

Best Practice: Keep analysis code versioned in Git. Store heavy `.rds` and `.h5` objects outside git tracking, backed up in `Data_backup`.

Cross-referencing biological cohorts across high-throughput sequencing, slide microscopy, QuPath digital pathology, and single-cell atlas objects.

COHORT ID	PRIMARY MARKERS / TARGET	INTEGRATED MODALITIES		ASSOCIATED WORKING DIRECTORIES	COMBINED SIZE
F2944	HEV vs. iBALT spatial microenvironment	Microscopy	QuPath	imaging_data_f2944_1, imaging_data_f2944_2, QuPath Analysis_F2944	1,579.07 GB
F2998 F3035	CD4 T cells, AdTx intervention cohorts	10x Single-Cell Seurat RDS		TB_TL_AdTx_F2998_F3035, TB_TL_TBD2_F2998, TB_TL_TBD3_F2998_F3035	345.73 GB
F2630	HEV structures & neutrophil infiltration	Microscopy	QuPath	imaging_data_f2630_HEV, imaging_data_f2630_neutrophils	517.37 GB
F2805	HEV maturation, B cell aggregates (wk 2, 4, 6)	Microscopy Spatial Graph		imaging_data_f2805_HEV, imaging_analysis_f2805_HEV	187.89 GB
Endothelial	CD31, CD86, sorted endothelial transcriptomes	Raw FASTQ	R/Seurat	sequencing_data_endothelial, TB-Endothelial, imaging_data_CD31pilot_#1_#2	754.84 GB
Total Lung	Whole-lung tissue atlas & spatial profiling	Raw FASTQ Atlas Repo	Microscopy	sequencing_data_totallung, TB-total-lung, tb-lung-atlas	1,035.44 GB

UCloud Storage Architecture, Backup Playbook & Governance

Operational runbook for mounting drives, automated Restic snapshot management, and FAIR compliance.

UCLOUD STORAGE TOPOLOGY & CONTAINER MOUNT NAMESPACE

DeiC Interactive HPC (K8s) • WekaFS 4.7 PB

/work/

TB group/

sequencing_data_*

imaging_data_*

NAFP Image analysis/

TB_TL_*

tb-lung-atlas/

Data_backup/

repo/

restic_pw.txt

JobParameters.json

job-report.csv

Container mount namespace (EPHEMERAL ROOT – Never store deliverables here!)

Primary Working Volume (Folder #1) – PERSISTENT (5.51 TB • 957,970 files)

Demultiplexed paired FASTQs (totallung, endothelial, monkey PAXgene)

Zeiss .czi & Leica .lif slides, QuPath caches, Cellpose training sets

QuPath multi-project annotations, measurements.csv & .qpdata

10x Genomics Cell Ranger count matrices & Scanpy AnnData suites

Version-controlled R/Seurat reproducible single-cell pipeline

Backup Repository Drive (Folder #2) – PERSISTENT (4.31 TB)

Automated Restic deduplicated snapshot repository

Secure password credentials for restic repository

Job execution parameters injected by UCloud K8s (ephemeral)

Continuous CPU, memory & I/O sampling telemetry (ephemeral)

Automated Restic Backup Playbook

Repository Location: /work/Data_backup/repo/ (always mount Data_backup as Folder #2).

Password File: Set export RESTIC_PASSWORD_FILE=/work/Data_backup/restic_pw.txt.

Exclusion Rules: Exclude temporary virtualenvs, checkpoints, and git caches:

--exclude=".venv" --exclude=".ipynb_checkpoints" --exclude="*.tmp" --exclude=".git/objects"

Snapshot Retention Policy: Enforce standard grandfather-father-son pruning:

restic forget --keep-daily 7 --keep-weekly 4 --keep-monthly 12 --prune

Integrity Check: Execute restic check --read-data-subset=5% monthly to verify repository chunk integrity.

FAIR Principles & Lab Data Hygiene

Findable (Naming Conventions): Maintain standardized directory prefixes:

sequencing_data_<cohort> for raw FASTQs.

imaging_data_<sample>_<stain/target> for microscopy.

TB_TL_<sample>_<subset> for single-cell count matrices.

Accessible: Access controlled via UCloud BINF INFIMM workspace membership. Files inherit POSIX group read permissions.

Interoperable: Raw data must remain in standard formats (FASTQ.GZ, CZI, LIF, H5, RDS). Tabular exports must use CSV/TSV with documented headers.

Reusable: Each analysis folder must contain a README.md describing sample metadata, experimental design, and pipeline version tags.

HPC Job Submission Golden Rules

Workspace Selection: Confirm BINF INFIMM is selected in the top-right workspace dropdown before launching any job.

Two-Folder Rule: Attach TB group as Folder #1 and Data_backup as Folder #2 on every compute run.

Scratch Usage: Perform heavy compilations, conda builds, and intermediate temp processing in /tmp; sync final deliverables to /work/TB group/.

Action Item Checklist for TB Team

[HIGH] Register 14 raw data directories in automated Restic backup schedule.

[HIGH] Audit 11 Git directories; commit unstaged code and push branches to GitHub.

[MED] Delete redundant clone QuPath Analysis F2944(1) after verification.

[INFO] Add missing README.md manifests in 17 directories lacking documentation.

INFIMM Bioinformatics Core • Department of Infection & Immunology • SDU eScience Center
Report compiled for: TB Research Group • Colleague Reference • Data Management

Source: inventory-tb-group.json • Git HEAD

✓ AUDIT VERIFIED

WekaFS /work/TB group/ • September 2026

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